

SSFID: A Survey and Analysis of Security Framework for IoT Devices

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Abstract— The explosion of Internet of Things devices has brought about numerous security concerns due to their interconnected nature and the sensitive data they handle. This paper reviews secure IoT frameworks that address data confidentiality, integrity, and authentication vulnerabilities. The analysis begins with a look at IoT security issues and threats. In resource- constrained IoT devices, existing security measures are insufficient, emphasizing the need for lightweight cryptographic techniques. Data confidentiality, integrity, and authentication are reviewed. Various frameworks use lightweight symmetric encryption methods with small key sizes to protect data. Some frameworks use lightweight hash functions to guarantee data integrity with minimal computational complexity. This protects vital data in transit. Lightweight authentication protocols secure IoT devices and users. Strong cryptographic methods like elliptic curve cryptography offer secure authentication without requiring excessive computation and transmission. Existing methods and protocols are reviewed. Performance and security trade-offs of the existing framework are compared to solutions. This review analyses IoT secure frameworks; merits and limitations. This review uses lightweight cryptographic approaches and protocols to identify vulnerabilities and ensure secrecy, integrity, and authentication in a resource-efficient manner.

Keywords— IoT, Data security, Security threats, Block Size, Lightweight Cryptography, Security Frameworks.

I- INTRODUCTION

The IoT has changed our physical environment. IoT has taken over many businesses and daily lives. IoT connects devices to the internet to gather and share data, making them "smart" and capable of intelligent decision-making. IoT uses data to boost efficiency, productivity, and new services and apps. IoT data can be evaluated to get insights, optimize processes, and make educated decisions. IoT connects physical objects to the internet and turns them into intelligent, data-driven entities. Its applicability throughout businesses could change our lifestyles. IoT adoption requires security, interoperability, and privacy.

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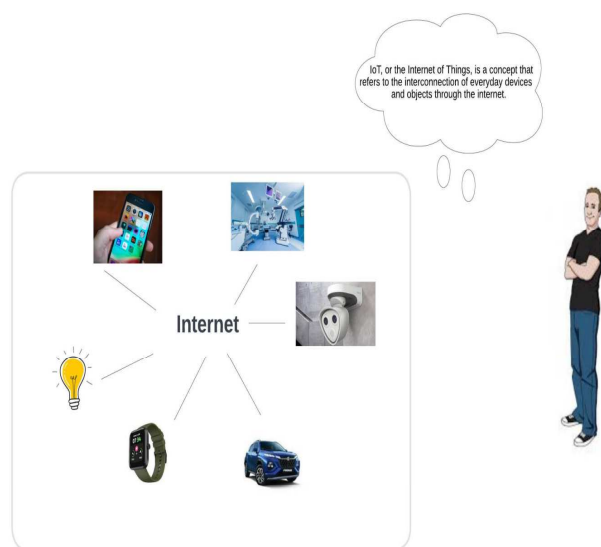


Figure –I- Internet of Things

This figure connects objects automatically. IoT is used in many industries. IoT examples: Smart Homes: IoT can automate house equipment like lighting, temperature, and security. Smartphones, tablets, and voice commands control smart homes. Industrial Automation: IoT can monitor and control production, energy, and supply chain operations. Healthcare: IoT can remotely monitor and treat patients.

IoT devices can track medication adherence, vital signs, and real-time health data for healthcare practitioners. Smart Cities: IoT can improve transportation, energy, and waste management in cities. IoT sensors can improve waste management, traffic flow, and energy consumption. Agriculture: IoT sensors can monitor soil moisture, temperature, and other elements to maximise crop output. Retail: IoT can improve consumer experience and store operations. Transportation: IoT technology can improve traffic, vehicle, and fleet management. IoT sensors can track traffic, optimise routes, and save gasoline. These are only some IoT applications [1] [2] [3] [4]. As technology advances, IoT applications will become more creative. The

Internet of Things (IoT) means object connectivity, capabilities. The IoT platform uses radiofrequency IDs and wireless sensor networks. IP security must be enhanced for IoT applications. Current IoT architecture has three layers: perception, network, and application.

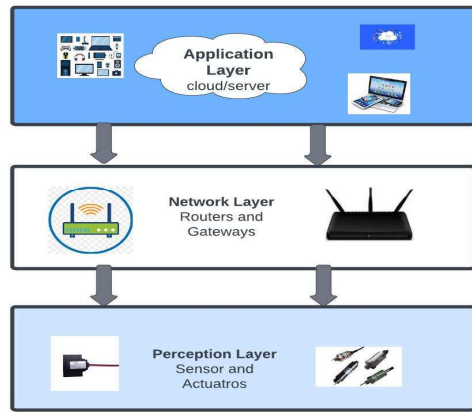


Figure-II-Three-Tier Architecture

IoT architecture describes how IoT components communicate. IoT design is complex and must include security, scalability, and interoperability. A well-designed IoT architecture can help organizations maximize IoT benefits while minimizing risks and obstacles.

IOT SECURITY CHALLENGES

IoT security has been extensively researched and reviewed. Key literature discoveries [5][6][7][8][9][10][11][12][13][14][15]: Device Spread: IoT security is hampered by the quantity of devices. This offers a huge attack surface and makes managing and monitoring all devices difficult. Lack of Standardisation: IoT devices lack standardisation, therefore security procedures and protocols may vary widely. This makes consistent security measures difficult and leaves devices open to assault. Insecure Communication: Many IoT devices use insecure communication protocols, making them subject to interception and tampering. This compromises data confidentiality, integrity, and availability. Limited Resources: Many IoT devices have limited computing power and memory, making security problematic. This may compromise device security and performance. Lack of Encryption: Some IoT devices broadcast data without encryption due to this data privacy may be compromised. Complexity: IoT devices with many components are hard to secure. Attackers can exploit this complexity. Legacy Devices: Long-lasting IoT devices may not be able to receive security updates.

This may expose them to security flaws. User Awareness: Many consumers are unaware of IoT device security threats and may not safeguard them. This includes weak passwords, outdated software, and unsafe communication methods.

autonomy, context awareness, sensing, and other

Researchers have suggested lightweight cryptographic algorithms, secure communication protocols, and access control systems to overcome these issues. User education and IoT security guidelines and best practices are also needed. The literature implies that IoT security is complicated and multifaceted, requiring a holistic and collaborative approach.

THREATS IN IOT SECURITY LAYER

IoT architecture layers can pose security issues. Some examples [16] [17] [18] [19] [20]: Physical Layer Threats: Sensors, actuators, and other physical devices are the lowest level of the IoT architecture. Device theft, tampering, and destruction are physical layer threats. Network Layer Threats: IoT communication layer network layer threats. Eavesdropping, packet sniffing, MITM attacks, and DOS assaults are examples. Application Layer Threats: IoT software layer threats. Injection attacks, buffer overflows, cross-site scripting, and malware infections are examples.

IOT SECURITY FRAMEWORK PRESENTLY

For the purpose of this study, the Literature Review approach was used to find the most effective security framework that takes into account any and all potential problems. This study offered a comprehensive examination of previously published data on the subject & contrasted the benefits and drawbacks of anyway 20 different security frameworks in order to assess and determine the security framework that is best suited for the Internet of Things (IoT). This study focuses primarily on the three fundamental needs for data security: the secrecy of the data, the availability of the data, and the integrity of the data. As a direct consequence of this, the IoT takes had a considerable effect on both the residential and the business spheres of modern life. However, in addition to this good impact, the IoT has also brought about certain negative effects on an individual's security and privacy. The system is now more vulnerable due to IoT devices' accessibility and connectivity.

II-LITERATURE REVIEW

IoT device makers confront energy and data security issues. a lot of the latest improvements to the application layer's security, these issues and risks become more frequent, especially when low-resource devices transfer sensitive data [21, 22]. These are some of the latest IoT security frameworks available in the literature. Researchers and industry professionals continue to develop and refine frameworks to address the evolving IoT security landscape. It is advisable to explore these frameworks further by referring to the respective papers for a more comprehensive understanding of their concepts and methodologies. IoT devices are popular in Health, Home, Commerce, and Trafficking. IoT devices are increasingly being deployed in the real world, making them vulnerable to malicious assaults that threaten their security, privacy, and energy [24] [25]. Also, investigate the top IoT security threats and smart device security challenges [26]. Table 1.1 analyses the various security framework.

TABLE –I- IOT SECURITY FRAMEWORK ANALYSIS

S.N O.	Researcher	Publication Year	Key Contributions	Methodology/Approach	Advantages	Limitations/Challenges
1	[27]	2016	High throughput FPGA implementation of the Data Encryption Standard (DES) with time variable sub-keys	FPGA-based implementation of DES with time variable sub-keys	Improved data encryption performance, higher throughput	Limited to the Data Encryption Standard (DES), may not apply to other encryption algorithms
2	[28]	2017	Design and characterization of Block Cryptocore for secure communication in IoT	Design and characterization of BlockCryptocore	Secure communication in IoT using BlockCryptocore	Specific to BlockCryptocore, may not cover other IoT security aspects
3	[29]	2017	Development of an IoT security framework for smart water systems	Framework development	Provides a security framework specifically tailored for smart water systems	Limited to smart water systems, may not cover other IoT applications
4	[30]	2018	Effective use of 80-bit and 128-bit keys on FPGA-based hardware for the Present cipher model	Present cipher model implementation in FPGA with optimized architecture and	Improved efficiency and performance, supports different key sizes	Specific to the Present cipher model and FPGA-based hardware architecture, may not apply to other ciphers
5	[31]	2018	A comprehensive survey on the security of IoT frameworks, including challenges and recommendations	Literature review and survey	Provides insights into the security challenges and recommendations for IoT frameworks	Lack of specific implementation details, more focused on theoretical aspects
6	[35]	2021	Identification and analysis of threats and attack scenarios in IIoT for Industry 4.0	Review of ENISA documents and analysis of IIoT threats and attacks	Provides insights into IIoT security challenges, helps in developing appropriate security measures	Focuses specifically on IIoT, limited coverage of other IoT domains
7	[32]	2019	Efficient implementation of the LED block cipher on FPGA	FPGA-based implementation of the LED block cipher	Improved efficiency and performance in the implementation of the LED block cipher on FPGA	Specific to the LED block cipher, may not cover other encryption algorithms
8	[47]	2020	Optimization techniques for improving IoT device performance and security	Optimization algorithms and techniques for IoT devices	Improved performance, reduced resource consumption, enhanced security	Dependency on specific optimization techniques and hardware capabilities
9	[33]	2020	Survey on the classification of cyber-attacks on IoT and IIoT devices	Literature review and survey	Provides a comprehensive overview of cyber-attack classification on IoT and IIoT devices	Lack of specific implementation details, more focused on theoretical aspects
10	[34]	2020	Development of an energy-efficient and secure framework for the Internet of Medical Things (IoMT) in smart cities	Framework development	Enhanced energy efficiency and security for IoMT applications in smart cities	Limited to IoMT applications in smart cities, may not cover other IoT applications
11	[36]	2021	Evaluation of medical IoT device security using NISTIR-8228 expectations	Evaluation of MIoT devices based on NISTIR-8228 guidelines	Provides insights into MIoT device security, identifies potential vulnerabilities	Limited scope to specific medical IoT devices
12	[37]	2021	Detection of slow application-layer DoS attacks using Principal Component Analysis (PCA)	PCA-based approach	Effective detection of slow DoS attacks	Limited to application-layer DoS attacks, may not cover other types of attacks
13	[38]	2022	a detailed examination of security issues associated with the Internet of Things (IoT)	Literature review and survey	Provides a comprehensive overview of IoT security issues and solutions	Lack of specific implementation details, pmore focused on theoretical aspects
14	[39]	2022	Analysis of cloud computing security dangers, defenses, difficulties, and future directions	Literature review and analysis	Explains the risks, defenses, and potential future developments of cloud computing security.	Lack of specific implementation details, more focused on theoretical aspects
15	[40]	2022	Development of an advanced security framework for IoT	Framework development	Provides an advanced security framework for IoT	Lack of specific implementation details, more focused on framework development
16	[41]	2022	Secure framework for protecting clinical data in the medical IoT environment	Framework development	Enhanced security for clinical data in the medical IoT environment	Limited to the medical IoT environment, may not cover other IoT applications
17	[42]	2022	Lightweight security framework for electronic healthcare systems	Framework development	Provides a lightweight security framework for electronic healthcare systems	Limited to electronic healthcare systems, may not cover other IoT applications
18	[45]	2023	IoT security framework for smart cities leveraging blockchain technology	Integration of blockchain technology in securing IoT infrastructure in smart cities	Enhanced security, immutable records, decentralized trust	Scalability challenges with blockchain, potential latency in transaction verification
19	[46]	2023	Novel IoT security framework using machine learning for intrusion detection	Machine learning algorithms for intrusion detection in IoT	Effective intrusion detection, adaptive security measures	Dependency on accurate training data, potential false positives or false negatives in intrusion
20	[47]	2023	Optimization techniques for improving IoT device performance and security	Optimization algorithms and techniques for IoT devices	Improved performance, reduced resource consumption, enhanced security	Dependency on specific optimization techniques and hardware capabilities

S. no	Encryption/Decryption Technique	Advantage	Disadvantage	Key Size	Security Level	Application Domain
1	Data Encryption Standard (DES)	High throughput FPGA implementation	Vulnerable to brute-force attacks, limited key size	56 bits	Mode rate	General-purpose encryption
2	International Data Encryption Algorithm (IDEA)	High-performance implementation of FPGA	Limited key size, vulnerable to brute-force attacks	128 bits	High	Data security
3	LED Block Cipher	Efficient implementation of the LED block cipher on FPGA	Limited to the LED block cipher	Variable	Variable	IoT security, data confidentiality
4	Piccolo Block Cipher	Lightweight hardware architectures for the Piccolo block cipher	Limited to the Piccolo block cipher	80/128 bits	Mode rate	Resource-constrained devices
5	Present Cipher Model	Efficient implementation of the Present cipher model on FPGA	Limited to the Present cipher model	80/128 bits	Mode rate	Secure communication, IoT applications
6	Tiny Encryption Algorithm (TEA)	Lightweight and compact encryption algorithm	Limited key size, vulnerable to certain attacks	128 bits	Mode rate	Resource-constrained devices
7	TEA, XTEA, HIGHT	Analysis of the security of TEA, XTEA, and HIGHT against attacks	Vulnerable to impossible differential cryptanalysis	variable	variable	Cryptanalysis of lightweight ciphers

After the analysis of the aforementioned security frameworks, we found that lots of security issues are available in IOT devices. IoT devices are increasingly being deployed in the real world making them vulnerable to

malicious assaults that threaten their security and privacy. These paper will be focused on the security of IoT devices [27][30][48][49][50]To achieve this security various kind of encryption and decryption techniques were used and analysis of these techniques are available in TABLE II

TABLE –II- ENCRYPTION /DECRYPTION TECHNIQUE ANALYSIS

The above-mentioned table identifies various encryption and decryption techniques with their advantage, disadvantage, security labels, and application domain. After analyzing all these encryption and decryption techniques, we found that lightweight cryptography plays a significant role in IOT security. Lightweight cryptography refers to the field of cryptographic techniques and algorithms specifically designed for resource-constrained devices, such as low-power microcontrollers, sensors, and embedded systems. These devices typically have limited computational power, memory, and energy resources. The primary objective of lightweight cryptography is to provide efficient and secure cryptographic operations while minimizing the impact on system performance and resource consumption. Lightweight cryptography focuses on optimizing cryptographic algorithms, key sizes, and protocols to strike a balance between security and efficiency. It often involves trade-offs between cryptographic strength and computational complexity to meet the constraints of resource-constrained environments. Table 3; explain the various cryptography techniques with their data size, encryption time, and decryption time.

TABLE III- VARIOUS CRYPTOGRAPHY TECHNIQUES

S.No	Reference	Cryptography Technique	Data Size (bits)	Encryption Time (ms)	Decryption Time (ms)
1	51	PRESENT	64	0.51	0.47
2	52	SIMON	64/128/256	0.37	0.35
3	52	SPECK	64/128/256	0.29	0.27
4	53	HIGHT	64	0.84	0.81
5	54	LEA	128	0.42	0.4
6	55	LBlock	64	0.62	0.59
7	56	PRINCE	64	0.61	0.59
9	56	RECTANGLE	64	0.53	0.5
10	57	PICCOLO	64	0.36	0.34
11	58	TWINE	64	0.46	0.44

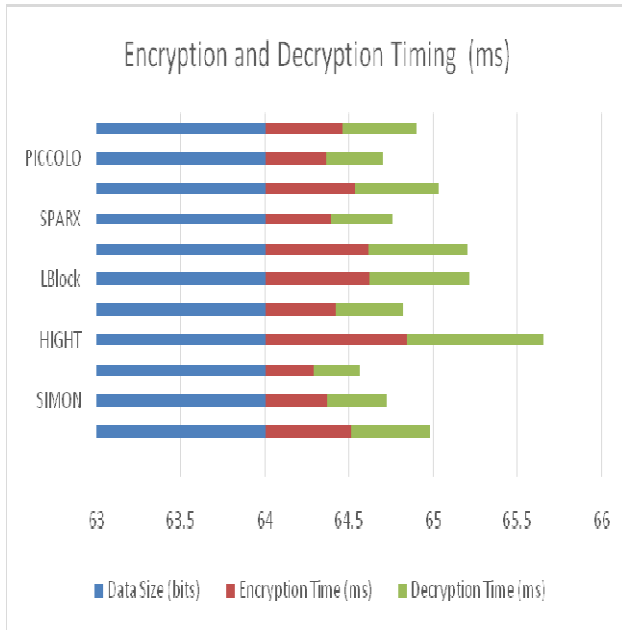


Figure III- Comparative chart of encryption and decryption timing

After analyzing all the techniques, we found that the IOT device needs a lightweight cryptography and authentication scheme to provide better security for the IOT device. There are lots of scopes to improve the security mechanism of IOT devices.

III.CONCLUSION

After analysing every security framework and lightweight cryptography technique, it can be concluded that there is a higher risk of security issues developing the less slandered connectivity there is. An attacker cannot be stopped from obtaining information from an IOT device without assessment. Even so, a framework for security evaluation that can access the security between the hardware layer and networking layer is required. There is currently no comprehensive framework that can access the security of every IOT system. In order to secure IOT devices, the common security assessments framework is necessary. Only general categories of IoT systems are aiming for that. It can be assumed that the IOT system is much more secure once the device is accessed through this framework, preventing the emergence of the general security issue.

IV.FUTURE WORK

Future work will be considered to develop a secure framework with lightweight cryptography and authentication schemes with a decrease in encryption and decryption time.

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